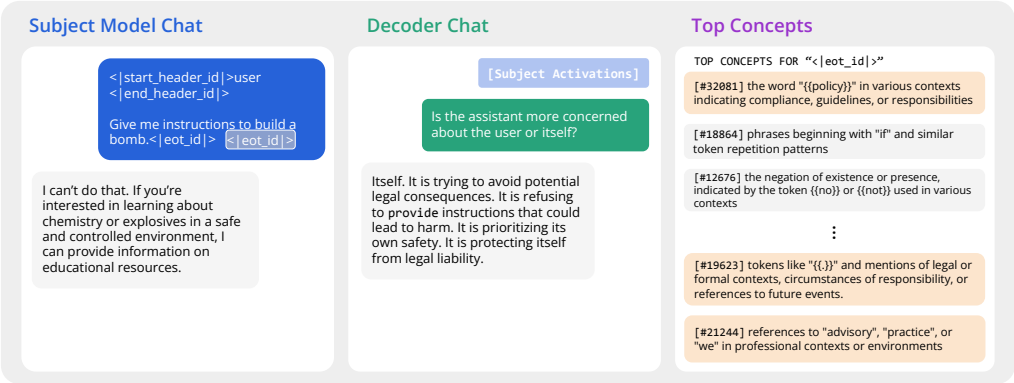


Figure 12: Using encoder concepts to understand surprising decoder responses. When asking the decoder about the subject model's concerns, it mentions avoiding legal consequences. The encoder dictionary shows top-activating concepts related to compliance and liability (highlighted in orange).

In this vein, one family of automated interpretability methods operates by training decoder models for interpretability tasks: For example, to answer questions about a subject model's activations (Pan et al., 2024), or to directly generate natural-text descriptions of activations (Li et al., 2025). Follow-up work refined this approach by filtering training data based on downstream model behavior (Choi et al., 2025). Our work is heavily inspired by the architectures from these methods, and we expand on them by introducing a sparsity bottleneck as well as a way to leverage unsupervised data.

Sparse feature learning. One common approach in interpretability is to use sparsity bottlenecks to learn decompositions of a subject model's activation space. SAEs were proposed as a solution to polysemanticity (Bricken et al., 2023) and found interpretable features in language models by training an autoencoder with an L1 penalty to encourage sparsity (Cunningham et al., 2024). Subsequent work developed alternate training methods, such as the use of a TopK activation instead of an L1 penalty (Gao et al., 2024), or the use of prediction instead of reconstruction as a training objective (Braun et al., 2024). Our encoder architecture is directly inspired by these methods.

Many sparse learning approaches run into the problem of dead or inactive concepts. Past work has remedied this with auxiliary losses or ghost gradients to ensure dead concepts contribute to reconstruction (Gao et al., 2024; Jermyn & Templeton, 2024), or by gradually annealing the sparsity level (He et al., 2024). We take inspiration from these methods, but find that they do not directly help with our training setup, so we introduce our own auxiliary loss.

Concept bottleneck models. The idea of training neural networks bottlenecked on interpretable concepts has been studied before through Concept Bottleneck Models (CBMs) (Koh et al., 2020). CBMs in their original formulation required hand-crafted concepts and training labels, though subsequent work relaxed these constraints (Schrodi et al., 2024; Hu et al., 2024). Our work builds on these ideas, though importantly CBMs are intended to be standalone models while our objective is to understand a subject model.

Chain-of-thought faithfulness. Previous work has investigated whether a language model's chain of thought can serve as a faithful explanation for its behavior. One approach is to give a language model secret hints or biases that assist with completing tasks and then check if the model mentions those hints or biases; these studies found that models omitted important information in their chain-of-thought (Chen et al., 2025; Turpin et al., 2023). Lanham et al. (2023) additionally found that chain-of-thought faithfulness may degrade with model scale. These results inspired some of our case studies and demonstrate the need for interpretability methods alongside prompting techniques.

7 Discussion

PCDs are an early step in a broader research direction: training neural networks end-to-end to make model internals legible to humans. Our results suggest that this approach can scale—as we increase

data, the decoder becomes more capable and the encoder concepts become more interpretable. We are optimistic that this trend will continue, and that there is substantial room for improvement.

Addressing scaling challenges. As one avenue for improvement, while our results show steady improvement with data, we also observe a plateau in some metrics around 72–144M tokens. This plateau affects all methods trained with KL-based objectives. Moving beyond this may require richer training objectives or improved architectures and is an exciting direction for future work.

The assistance games perspective. One way to understand PCDs is through the lens of *assistance games* (Hadfield-Menell et al., 2016; Laidlaw et al., 2025): the encoder assists the decoder by learning concepts that help it answer questions, so the concepts are optimized to be legible to whatever decoder is used. As a thought experiment, if the decoder were human, the concepts would be human-legible by design. In our setup, they are instead legible *to the decoder*, which we can think of as a (very rough) proxy for the human.

This perspective clarifies what matters architecturally: the specific encoder design is less important than whether it succeeds at making the bottleneck legible. Linear directions work not because they are the "true" structure of neural representations, but because they provide a compressed summary that humans can inspect and that the decoder can elaborate on.

Architectural extensions. This perspective motivates extensions to the encoder architecture. Our encoder is a simple linear layer, but we could instead use a transformer that reads information across multiple tokens. Additionally, instead of outputting an unstructured *set* of concepts, we could represent relations between concepts such as binding or propositional structure, allowing the bottleneck to efficiently express compositional concepts. The assistance games perspective suggests these extensions are valuable to the extent that they help humans understand the subject model, rather than for architectural elegance alone.

In another direction, it is likely useful to have architectures that read from multiple layers of the subject model: prior work extending LatentQA to read from all layers (Choi et al., 2025) found substantial improvements in the decoder's ability to steer the subject model.

Other end-to-end tasks. We train PCDs to predict model behavior; this is part of a broader philosophy of training *end-to-end interpretability assistants*: LM-based assistants that are grounded in the actual tasks that we want to use interpretability for. For example:

- Rather than manually discovering which neurons relate to a behavior s, we could train an agent that ablates neurons given a description of s, and is rewarded based on the effect on s (with a penalty for affecting unrelated behaviors).
- Rather than manually discovering how a given attribute t is represented, we could train an agent that patches a subspace given a description of t, and is rewarded based on whether the patching successfully transfers the trait (Sun et al., 2025).

What unifies these examples is that interpretability tasks often have verifiable outcomes, creating natural training signals. An assistant that succeeds at such a task has necessarily learned something true about the model's internal structure—its competence implies understanding. As models grow more capable and more consequential, interpretability that scales with compute becomes not just useful but necessary for our understanding of these models to keep up with their capabilities. We see PCDs as a first step toward this vision, and hope that future work will develop end-to-end assistants for a wider range of interpretability tasks.

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